Day 4 Task Report

Title: Prompt Engineering

TASK-04: Resolving Model Hallucinations via Structured Prompting.

Objective

The objective of this task is to understand how Large Language Models (LLMs) can produce incorrect or hallucinated outputs due to poorly structured prompts. In this task, we fix a broken prompt using techniques like:

- Few-Shot Prompting
- Structured Output Formatting
- Anti-Prompts (Negative Constraints)

This helps ensure the model produces accurate and reliable results.

Tools Used

- VS Code
- Git & GitHub
- MS Word

Libraries Used

- **google-genai**: To interact with Gemini API
- **dotenv**: To securely store API key
- **json**: To handle structured output
- **PyPDF2**: To read resume from PDF file

Concepts Used

1. Prompt Engineering

Designing structured prompts to control model behavior and reduce hallucination.

2. Few-Shot Prompting

Providing examples in the prompt so the model learns expected input-output format.

3. Output Constraints

Forcing the model to return output strictly in JSON format.

4. Anti-Prompts (Negative Constraints)

Preventing the model from making assumptions like:

- "Do not assume Python knowledge"

5. Hallucination Reduction

Ensuring the model only uses information explicitly present in the resume.

Code Explanation

The program works in two modes:

1. Text Input Mode

User directly enters resume text.

2. File Input Mode

User provides a PDF file → extracted using PyPDF2.

Then:

- Prompt + Resume → sent to Gemini API
- Model returns structured JSON
- JSON is parsed and printed

Errors

1. Deprecated Library Issue
Old library google.generativeai stopped working

Solution:
Used updated library using “google.genai”

2. Model Not Found Error

Solution:
Updated model name to: “gemini-flash-latest”

3. Quota Exceeded Error

429 RESOURCE_EXHAUSTED

Solution:

- Wait for quota reset
- Ensure API key is properly configured

4. Missing PyPDF2

Solution:

Installed package (`pip install PyPDF2`).

Output

```
}
PS D:\Qualtech\Daily Tasks\Day4> py -3.13 resume_task.py
Enter your resume text: Name:Shweta Patil, I have 2 yrs of experience at Techcorp and worked on Java, c#, and SQL database.
{
  "name": "Shweta Patil",
  "skills": [
    "Java",
    "C#",
    "SQL database"
  ],
  "python_present": false,
  "fit_for_python_role": "no",
  "reason": "Python not mentioned",
  "eligibility": "Not Eligible"
}
PS D:\Qualtech\Daily Tasks\Day4> py -3.13 resume_task.py
Enter 1 for text input or 2 for file input: 1
Enter your resume text: Karan Sankapal, I have completed my diploma in electronics and I known languages like C,C++,Java
{
  "name": "Karan Sankapal",
  "skills": [
    "C",
    "C++",
    "Java"
  ],
  "python_present": false,
  "fit_for_python_role": "no",
  "reason": "Python not mentioned",
  "eligibility": "Not Eligible"
}
```

```
PS D:\Qualtech\Daily Tasks\Day4> py -3.13 resume_task.py
Enter 1 for text input or 2 for file input: 2
Enter PDF file path: Riya-Resume.pdf
{
  "name": "Riya Bhosale",
  "skills": [
    "C",
    "C++",
    "Java",
    "Python (Numpy, Pandas, Matplotlib)",
    "SQL",
    "PostgreSQL",
    "My SQL",
    "Supervised models",
    "Unsupervised models",
    "Power BI",
    "Git",
    "GitHub",
    "Jupyter Notebook",
    "VS Code"
  ],
  "python_present": true,
  "fit_for_python_role": "yes",
  "reason": "Python is mentioned",
  "eligibility": "Eligible"
}
```

Conclusion

This task helped me understand how important prompt design is while working with AI models.
I learned how to:

- Reduce hallucinations using structured prompts
- Control output using constraints
- Use few-shot learning effectively
- Handle real-world API issues like quota and errors